Exercise A.6

Given a network which is monitored by a three IDS with beliefs

Sensor A

Decision	mass
Attack	0.3
Safe	0.2
Don't know	0.5

Sensor B

Decision	mass
Attack	0.35
Safe	0.25
Don't know	0.4

Sensor C

Decision	mass
Attack	0.35
Safe	0.1
Don't know	0.55

{A,S}

what is the likelihood that the network is under attack?

$m(A) = \sum_{S_i \cap S_j = A} m(S_i) \cdot m(S_j)$

A+B	m
{A}	$0.3 \cdot 0.4 + 0.3 \cdot 0.35 + 0.35 \cdot 0.5 = 0.4$
{S}	$0.2 \cdot 0.4 + 0.25 \cdot 0.5 + 0.2 \cdot 0.25 = 0.2550$
{A,S}	$0.5 \cdot 0.4 = 0.2$

$K = 0.4 + 0.2550 + 0.2 = 0.8550$

$m_2(A) = 0.4678$, $m_2(S) = 0.2982$, $m(A,S) = 0.2339$

A+B+C	m
{A}	$0.4678 \cdot 0.35 + 0.4678 \cdot 0.55 + 0.35 \cdot 0.2339 = 0.5029$
{S}	$0.2982 \cdot 0.1 + 0.2982 \cdot 0.55 + 0.1 \cdot 0.2339 = 0.2172$
{A,S}	$0.2339 \cdot 0.55 = 0.1286$

$K = 0.8497$

$m_3(A) = 0.5926$, $m_3(S) = 0.2559$, $m_3(A,S) = 0.1515$

Exercise A.5

Given a system failure, two experts (John and Paul) are evaluating the different HIDS and NIDS of different machines to identify the source of the attack. System failure could be caused be Terminal A, Terminal B or Terminal C.

John believes that the failure is due to terminal A with a mass m1(A) = 0.8 or to terminal B with mass m1(B) = 0.1; uncertainty is 0.1.

Paul believes that the failure is due to computer C with m2(C) = 0.9 or to the others with mass 0.05 (m2(A,B)); uncertainty is 0.05.

Compute a Dempster-Schafer combination of John and Paul opinions.
Describe each step and make a table of values.

John	m
{A}	0.8
{B}	0.1
{A,B,C}	0.1

Paul	m
C	0.9
{A,B}	0.05
{A,B,C}	0.05

J+P	m
	0.4211
{A}	0.8 · 0.05 + 0.8 · 0.05 = 0.08
{B}	0.1 · 0.05 + 0.1 · 0.05 = 0.01 0.0526
{C}	0.9 · 0.1 = 0.09 0.4737
{A,B}	0.05 · 0.1 = 0.005 0.0263
{A,B,C}	0.1 · 0.05 = ~~0.005~~ 0.0263